

## DEEPAK BHATTARAI

### AI/ML Research | Software Engineering

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## RESEARCH PROFILE

MSc Artificial Intelligence graduate with Distinction and 9+ years of software-development experience, combining applied AI research with production-oriented software engineering. Research experience spans deep-learning video classification, computer vision, temporal modelling, model evaluation, anomaly detection and real-time inference. Applied engineering work includes LLM-enabled assessment systems, audio workflows, full-stack API platforms and cloud/VPS deployment. Research interests include agentic AI, AI-assisted software engineering, trustworthy AI, computer vision and real-time intelligent systems.

## RESEARCH INTERESTS

Agentic AI & AI-assisted software engineering • Trustworthy & interpretable AI • Computer vision • Deep learning & temporal modelling • Real-time intelligent systems • Applied machine learning

## EDUCATION

**MSc Artificial Intelligence - Distinction** | London Metropolitan University | UK **2024-2025**

- Dissertation: Deep Learning-Based Driver Monitoring System for Real-Time Detection of Aggressive, Drowsy, Distracted and Normal Driving Behaviours.

**Bachelor's degree in computer engineering - 60%** | Kantipur Engineering College | Nepal **2019**

## MSc DISSERTATION RESEARCH

**Real-Time Driver Monitoring System** | Deep Learning / Computer Vision Research

- Designed a real-time monocular-RGB, four-class driver-monitoring system using balanced 8-second video clips to classify Normal, Aggressive, Distracted and Drowsy driving behaviours.
- Built and compared lightweight spatiotemporal deep-learning pipelines using a MobileNetV2 CNN backbone with temporal architectures including bidirectional LSTM, Transformer and temporal-convolution approaches; implemented training and preprocessing in Python with TensorFlow/Keras and OpenCV.
- Evaluated models using accuracy, precision, recall, F1, macro ROC-AUC, mAP, calibration/ECE and confusion-style error analysis; investigated class imbalance, dominant-class behaviour, misclassification and prediction stability.
- Developed a rolling-buffer real-time inference prototype with frame sampling, RGB normalisation/contrast enhancement, temporal smoothing and latency/FPS testing, with an emphasis on efficient deployment on resource-constrained hardware.

## SELECTED AI & TECHNICAL PROJECTS

**Freelance AI & Full-Stack Developer - Language Vision** | English Language Learning Platform | **2026-Present**  
UK / Remote

- Architected and developed an AI-enabled English-learning and assessment platform using FastAPI, SQLAlchemy, Pydantic, Alembic, PostgreSQL, Next.js and TypeScript, with JWT authentication and role-based admin, teacher and student workflows.
- Integrated LLM APIs for AI-assisted writing and speaking evaluation, structured feedback and assessment workflows, alongside deterministic/rule-based marking for objective question types.
- Designed listening and speaking workflows with audio prompts, browser recording, multi-question audio sets and real-time/live-class components using LiveKit/WebRTC-oriented infrastructure.
- Built database models, migrations, APIs, import/assessment pipelines and production deployment workflows using Docker/VPS hosting, object storage and CI-style Git/GitHub practices; debugged performance, session and data-integrity issues across a multi-user production system.

**Smart Environmental Monitoring & Anomaly Detection** | AI / IoT Project

- Built an IoT environmental-monitoring pipeline using Raspberry Pi/Arduino sensors, Python and Azure IoT, with live data aggregation, dashboard visualisation and anomaly-detection objectives.

**Nepal Disaster Relief Platform** | Full-Stack / Data Automation | [reliefnepal.com](#) **2026-Present**

- Developed and stabilised a Next.js/FastAPI/PostgreSQL disaster-information platform with disaster archives, Situation Room, missing-person and funding/transparency views; hardened official-data workers/APIs using polling, caching, timeouts and stale-value preservation, improving pagination, performance, logging and regression testing.

## TECHNICAL RESEARCH & ENGINEERING SKILLS

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|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AI / ML</b>                   | Machine learning, deep learning, computer vision, transfer learning, temporal modelling, model evaluation, anomaly detection, LLM API integration, real-time inference |
| <b>Programming</b>               | Python, JavaScript, TypeScript, PHP, Dart, C, C++, SQL                                                                                                                 |
| <b>AI / Backend</b>              | TensorFlow/Keras, OpenCV, FastAPI, SQLAlchemy, Pydantic, Alembic, Node.js, Express.js, Laravel, REST APIs, JWT authentication                                          |
| <b>Frontend / Mobile</b>         | React, Next.js, React Native, Flutter, Redux/state management, HTML5, CSS3, Tailwind CSS, Bootstrap                                                                    |
| <b>Data / Real-time / Cloud</b>  | PostgreSQL, MySQL, SQL Server, MongoDB, Firebase, Azure IoT, LiveKit/WebRTC workflows, DigitalOcean object storage                                                     |
| <b>Infrastructure / Delivery</b> | Docker, VPS deployment, Dokploy-style container deployment, Git/GitHub, API integration, performance optimisation, debugging                                           |
| <b>Engineering Practice</b>      | Software architecture, database/schema design, unit & integration testing, code review, Agile/Scrum, Jira, Trello, mentoring                                           |

## PROFESSIONAL EXPERIENCE

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- PHP Developer** | School of Dental Nursing | UK **Nov 2024 - May 2026**
- Developed and maintained PHP/Laravel learning and administration systems covering courses, student records, exams, dashboards, payments, reporting, authentication, role-based access, email notifications and third-party APIs.
  - Improved backend logic, database queries, validation, file/document workflows, performance and production reliability while providing technical support to operational users.
- Mid-Level Developer** | Omega BPO Outsourcing Pvt Ltd | Nepal **Apr 2023 - Aug 2024**
- Delivered production web and mobile features with React, Next.js, Flutter and React Native; integrated backend APIs, created reusable components and contributed to architecture and technical design decisions.
  - Mentored junior developers, reviewed code and helped maintain engineering standards across complex team projects.
- Junior Software Developer** | Bug Byte Technology Pvt Ltd | Nepal **Oct 2021 - Mar 2023**
- Worked across requirements, design, implementation, testing and maintenance; designed relational/NoSQL schemas, SQL queries, stored procedures and application data flows.
- React Native Developer** | Progressive Labs Pvt Ltd | Nepal **Nov 2020 - Sep 2021**
- Developed cross-platform iOS/Android features, integrated APIs, applied Redux/state-management patterns, debugged application issues and contributed to unit-testing and code-quality practices.
- Software / React Native Developer** | SPG Technologies | New Delhi, India **Feb 2017 - Sep 2020**
- Built responsive web and mobile applications with JavaScript, React and React Native; collaborated with product, business and QA teams across requirements, delivery, testing, optimisation and maintenance.

## ADDITIONAL PROJECT EXPERIENCE

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- Jodinee.com** - Flutter/Laravel matrimony platform with Firebase, Dio and Provider; implemented authentication/onboarding, profile/discovery, interests, matches/chat, push notifications, media uploads and deployment workflows.
- NepalUK.com** - Laravel-based e-commerce/community web platform involving backend application development, database workflows and user-facing features.
- Wizam.com** - NEBDN Online Exam Platform | Laravel, PHP, Next.js, React, MySQL, Stripe - Developed and maintained mock-test and examination workflows, subscription and one-off payments, exam access control, admin management, transactional emails, FAQs, and production bug fixes/improvements.